

Animate Subject Tracks — User Guide

This workflow turns an EarthRanger subject group's GPS history into an animated 3-D map: colored trails that sweep across real terrain as time passes, with an optional exportable video. It's built and run through the **Ecoscope Workflows** platform — no coding required to use it.

1. What this workflow does

For a subject group you choose (e.g. "Elephants"), the workflow:

1. Connects to your EarthRanger site and pulls location fixes for every subject in the group over a time range you set.
2. Cleans the data and converts it into movement trajectories.
3. Renders those trajectories as an animated, terrain-aware 3-D map — each subject gets its own colored trail with a moving "head" marker.
4. Optionally renders that animation to a downloadable `mp4` video.
5. Packages everything into a dashboard you can view, share, or revisit later.

2. Before you start

You'll need:

- Access to the **Ecoscope Workflows** platform.
- An **EarthRanger connection** already configured on the platform (ask your Ecoscope admin if you don't see one in the dropdown).
- The exact **subject group name** as it appears in EarthRanger (e.g. `Elephants`) — subject groups are managed in EarthRanger, not in this workflow.
- A rough idea of the **date range** you want to animate. Longer ranges take longer to fetch and render.

3. Running the workflow

1. Open the Ecoscope Workflows platform and select **Animate Subject Tracks** from the workflow catalog.
2. Fill in the form (see field-by-field guide below).
3. Click **Run**.
4. Wait for the run to complete — you'll see a status indicator while the workflow fetches data and renders the map. Enabling video export adds noticeably more time, since the animation is re-rendered frame-by-frame.
5. Open the resulting **dashboard** to view your animated map (and video, if you enabled it).

4. Filling in the form

The form is organized into sections, top to bottom. Most fields have sensible defaults — the only things you truly need to set are the **data source**, **time range**, and **subject group name**. Everything under an "Advanced" / optional section is safe to leave alone on your first run.

Set workflow details

Field	Required?	Notes
Workflow Name	Yes	A label for this run, e.g. <code>Elephants - July 2026</code> . Helps you find it again later in your run history.
Workflow Description	No	Free-text notes about this run.

Connect to EarthRanger

Field	Required?	Notes
Data Source	Yes	Choose the EarthRanger site/connection to pull data from.

Define analysis time range

Field	Required?	Notes
Since	Yes	Start of the period to animate.
Until	Yes	End of the period to animate.

Timezone	No	Interpret Since/Until in a specific timezone.
Time Format (advanced)	No	Only affects how times are displayed elsewhere in the dashboard.

Tip: the wider the range, the more GPS fixes get pulled and animated — which means a longer run and a longer video. For a first try, a window of a few days to a couple of weeks is a good size.

Subject Group

Field	Required?	Notes
Subject Group Name	Yes (default: <code>Elephants</code>)	Must exactly match a subject group name configured in EarthRanger. This determines which subjects' tracks get animated.

Convert relocations to trajectories (advanced, optional)

A **Trajectory Segment Filter** removes GPS noise by discarding movement segments that are too short, too long, too fast, or too slow to be real animal movement (e.g. a GPS glitch that implies teleporting 100 km in one second). The defaults work well for most terrestrial wildlife — you generally don't need to touch this unless you're animating a species with very unusual movement patterns (e.g. birds).

Configure terrain elevation decoder (advanced, optional)

- **Exaggeration** — stretches terrain height for visual effect. `1.0` true-to-scale, `2.0` doubles apparent elevation, useful for making subtle hills/valleys more visible on flat landscapes.

Calculate map view bounds (advanced, optional)

- **Pitch** — camera tilt, `0` (top-down, default) to `90` (looking at the horizon).
- **Bearing** — compass rotation of the map, `-180` to `180`, `0` = north up.

Configure animation settings (advanced, optional)

- **Animation Speed** — how fast simulated time passes per real second of playback. Higher = faster playback.
- **Head Radius / Head Outline Width** — size and outline thickness of the moving dot that marks each subject's current position.

Draw animated map (*advanced, optional*)

- **Head Layer** — swaps the flat colored dot for a 3-D animal model (a bundled elephant model by default) that rotates to face its direction of travel. Off by default. Turn on **Enabled** to use it; the other fields here (size, model orientation, etc.) fine-tune how that 3-D model looks and behaves and can be left at their defaults.

Video Creation (*optional — off by default*)

Turn this on to also render the animation as a downloadable  in addition to the interactive map.

Field	Notes
Enabled	Turns video rendering on. Leave off if you only want the interactive map (faster run).
Camera	How the video's camera moves. Choose one:

- **Static** — camera holds one fixed view for the whole video.
- **Fit** — camera zooms out just enough to keep every point visited so far in frame.
- **Follow** — camera tracks a subject (or the group) from directly above.
- **Follow 3D** — like Follow, but tilted for a 3-D chase-cam feel, rotating to match the subject's heading.
- **Orbit** — camera circles slowly around the center of all the tracks.
- **Cinematic** — a smooth fly-through that leads the subject and banks with its turns.
- **Keyframes** — camera flies through a custom path, either auto-derived by following a chosen subject or built from an uploaded waypoint file.

5. Understanding your results

When the run finishes, open the dashboard. You'll see:

- **Subject movements** — the interactive animated 3-D map. Use the builtin playback controls to play, pause, and scrub through the time range. Each subject appears as its own colored trail with a legend on the map listing subject names and colors.

- **Video download** (*only if Video Creation was enabled*) — an image of the same animation, suitable for sharing or presentations.

The dashboard also records the workflow name/description, the time range, and the subject group you selected, so you (or a colleague) can tell at a glance what a given run covers.

6. Troubleshooting

Symptom	Likely cause
The map is empty / no trails appear	The subject group name doesn't match EarthRanger exactly, or there are no location fixes for that group in the chosen time range.
Run takes a long time	The time range is very wide, the subject group is large, or Video Creation is enabled — all three multiply how much data is fetched and rendered.
A subject's trail looks broken into short disconnected pieces	The Trajectory Segment Filter (advanced) may be discarding segments as noise. Only adjust this if you understand your data's typical fix intervals and speeds.
The 3-D animal model doesn't appear	The Head Layer toggle under "Draw animated map" is off by default — enable it to render 3-D models instead of flat dots.